

Quiz 1

1. Which sequence equals $\delta[n - 3]$?

A. $u[n - 3] - u[n - 4]$, B. $u[n + 3] - u[n + 2]$, C. $u[n - 2] - u[n - 3]$, D. $u[n] - u[n - 3]$

Solution: A

2. The system $y[n] = x[n - 2] + x[n + 2]$ is:

A. linear, time invariant, causal B. linear, time invariant, noncausal C. nonlinear, time invariant D. linear, time varying

Solution: B

3. Let $x[n] = \{1, 2\}$ and $h[n] = \{1, -1\}$, both beginning at $n = 0$. Compute $y[n] = x * h$.

$y[-1] =$ $y[0] =$ $y[1] =$ $y[2] =$ $y[3] =$

Solution: $y[-1] = 0$ $y[0] = 1$ $y[1] = 1$ $y[2] = -2$ $y[3] = 0$

4. Why are complex exponentials called eigenfunctions of LTI systems?

Answer in one or two sentences.

Answer: Complex exponentials pass through LTI systems unchanged in shape. Only amplitude and phase changes.

5. If $X(e^{j\omega})$ is a DTFT, what is $X(e^{j(\omega+2\pi)})$?

Explain why in one line.

Solution: $X(e^{j(\omega+2\pi)}) = X(e^{j\omega})$. It is the same DTFT because $X(e^{j\omega})$ is periodic in ω with period 2π .

6. For $y[n] = 0.5y[n - 1] + x[n]$, determine $H(e^{j\omega})$ at $\omega = 0$ and $\omega = \pi$.

$H(e^{j0}) =$ $H(e^{j\pi}) =$

Solution: $H(e^{j0}) = 2$ $H(e^{j\pi}) = \frac{2}{3} = 0.667$

